Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a qubit spectroscopy method based on Lorentzian-derived drive pulses that enhances robustness against amplitude variations. Specifically, we employ a root-Lorentzian envelope with a π-phase inversion at its peak, which introduces an echo-like symmetry that cancels on-resonant Rabi oscillations while preserving the spectral response. This approach suppresses conventional power broadening and is compatible with driven-spectroscopy pulses shorter than the qubit lifetime T_1, a regime in which square and ordinary root-Lorentzian pulses retain amplitude-dependent coherent oscillations. Numerical simulations predict a narrow echo feature over more than three decades of peak drive. Within the independently selected experimental window of approximately 2.5–9 MHz, the fitted resonance center exhibits a 3 kHz RMS displacement and a 7.5 kHz maximum excursion, without the monotonic AC-Stark shift expected for a constant drive. We experimentally demonstrate this method on a superconducting qubit platform and observe qualitative agreement with numerical simulations, highlighting its potential for calibration-robust qubit frequency characterization as qubit lifetimes increase.

Accurate frequency characterization of a quantum two-level system (qubit) is essential for calibrating high-fidelity control in fault-tolerant quantum processors [1]. In many physical implementations, such as superconducting circuits and semiconductor quantum dots, the qubit is realized within an inherently multilevel structure with discrete energy levels, allowing two levels to be selected to form an effective computational subspace that mimics atomic systems via their quantized energy spectrum [2, 3]. Spectroscopic identification of the transition frequency is therefore routinely performed by applying resonant control fields and monitoring the resulting population dynamics [4–6]. Spectroscopic resolution in direct population spectroscopy of coherently driven two-level systems is constrained by decoherence. In the absence of technical noise, stochastic phase fluctuations induce homogeneous broadening of the transition frequency, leading to a minimum resolvable linewidth that scales inversely with the coherence time T_2, as described by the Bloch equations [7, 8]. This establishes the so-called T_2 limit for conventional direct spectroscopy.

In practice, however, driven spectroscopy introduces an additional trade-off between signal robustness and spectral resolution. When finite-amplitude control pulses are applied, the system dynamics deviate from the idealized two-level response due to coherent Rabi oscillations and coupling to noncomputational levels [3, 9, 10]. Increasing the drive amplitude improves signal contrast but simultaneously induces power broadening, resulting in a linewidth that scales with the instantaneous Rabi frequency [11]. Consequently, conventional square-pulse and continuous-wave spectroscopic techniques must operate at low drive amplitudes to maintain resolution approaching the decoherence-limited linewidth set by T_2, creating a severe practical trade-off between precision and robustness. Smooth temporal envelopes can weaken this broadening, and powers-of-Lorentzian pulses have been shown theoretically and experimentally to exhibit power narrowing [12, 13].

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit [14]. Achieving such resolution, however, requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Finite-pulse imperfections and probe-induced frequency shifts, including unintended AC Stark shifts during the control pulses, can nevertheless bias the extracted transition frequency [15, 16].

Alternative protocols can evade a probe-T_2 resolution bound by using inter-shot phase correlations, external clock references, dynamical decoupling, or tailored filter functions [17–22]. These approaches address different sensing settings and typically require repeated correlations, extended interrogation times, a stable external reference, or additional control resources. Here, we address the resolution–robustness trade-off within direct driven population spectroscopy by employing adiabatically engineered Lorentzian-derived drive pulses with a π-phase inversion at their peak. Unlike prior Lorentzian-pulse power-narrowing demonstrations, this echo-like structure suppresses on-resonant Rabi oscillations while preserving the off-resonant spectral response, maintaining narrow frequency features at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude.

Here, we propose a spectroscopic method for probing the qubit transition frequency using arbitrarily shaped drive pulses, specifically an echo-root-Lorentzian pulse. Previous theoretical work showed that Lorentzian pulses exhibit power narrowing, complementary to the conventional power broadening observed for constant drives, as a consequence of their adiabatic properties [12]. Power narrowing with powers-of-Lorentzian pulses was subsequently demonstrated experimentally on IBM Quantum,

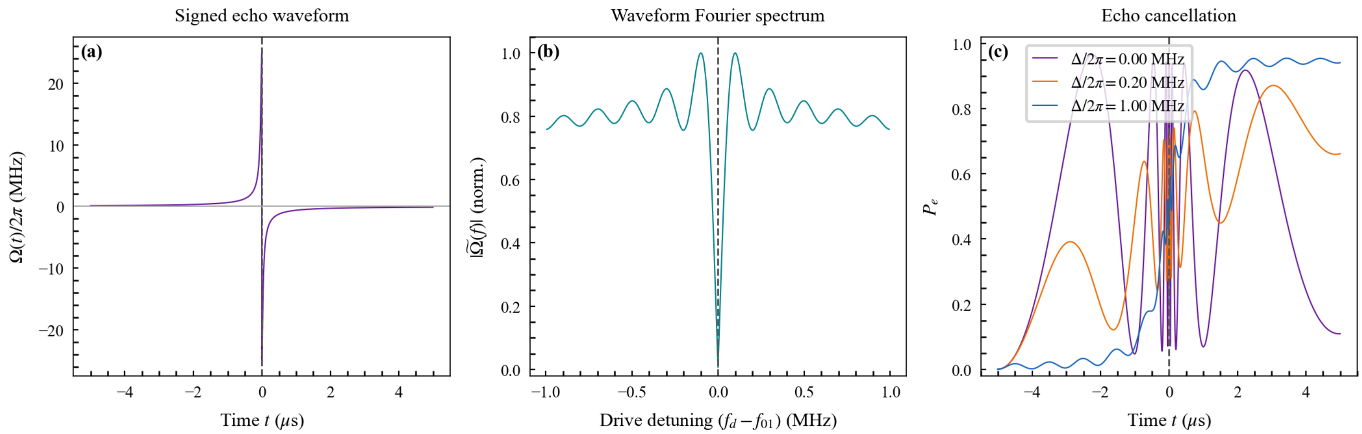

Figure 1. Pulse waveform and echo-cancellation mechanism. (a) Signed echo-root-Lorentzian drive envelope with a π phase inversion at the pulse midpoint. (b) Normalized magnitude of the waveform Fourier spectrum, showing the spectral null at zero frequency produced by the phase inversion. (c) Simulated excited-state population during the pulse at resonance and at non-resonant detunings of $\Delta/2\pi = 0.20$ and 1.00 MHz, using $T_1 = 9999$ μs. At resonance, the evolution accumulated during the first half is reversed during the second half, whereas finite detuning prevents complete cancellation.

including the cutoff broadening introduced by truncating the pulse wings [13]. In this work, we extend this framework to longer pulse durations approaching the T_1 timescale and introduce a π-phase jump at the pulse peak. The resulting echo property effectively cancels the net pulse area while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the echo-root-Lorentzian drive from the order-n generalized Lorentzian envelope as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \tag{1}$$

$$L^{(1/2)}_{\text{echo}}(t) := L^{(1/2)}(t)\,\mathrm{sgn}(t). \tag{2}$$

where τ sets the characteristic pulse duration and n controls the algebraic decay of the wings. Throughout the theory, $\Delta = \omega_d - \omega_{01}$ is the drive–qubit detuning, and Δ and Ω are angular frequencies. Plotted axes are labeled $\Delta/(2\pi)$ and $\Omega_0/(2\pi)$. We denote an angular-frequency full width at half maximum by Γ_ω and its value in hertz by $\Gamma_f = \Gamma_\omega/(2\pi)$.

For the untruncated, plain generalized Lorentzian and $n > 1/2$, the usual adiabatic-breakdown estimate gives [12]

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-\frac{1}{2n-1}}, \tag{3}$$

where Δ_b is the angular detuning at which adiabatic following ceases and Ω_0 is the peak angular Rabi frequency. The exponent grows without bound as $n \to 1/2^+$, so within its domain this asymptotic result predicts the steepest power-narrowing trend at the root-Lorentzian boundary. Exactly at $n = 1/2$, however, Eq. (3) is singular and cannot be evaluated: it does not predict a zero linewidth or an infinite resolution. Finite truncation, decoherence, and the echo phase inversion regularize the response, and the FWHM Γ_f used below is therefore obtained from the complete simulated or measured spectrum. This interpretation is consistent with the power narrowing and cutoff broadening observed for powers of Lorentzian pulses by Mihov and Vitanov [13]. In contrast, a square pulse exhibits conventional power broadening, with $\Gamma_\omega \propto \Omega_0$ at large drive. The derivation and the marginal $n = 1/2$ limit are detailed in the Supplemental Material [23]. Numerical spectroscopy maps for matched root-Lorentzian and echo-root-Lorentzian pulses are shown at broad and narrow detuning scales in Fig. 2. All three rows use $L = 20$ μs; the Lorentzian-derived pulses use $c = 0.002$, and the columns span 50 and 1 MHz, respectively, with matched model parameters. Independent measurements on qubit q1 give $T_1 = 51.24 \pm 0.73$ μs and a Ramsey time $T_2^* = 7.31 \pm 0.13$ μs. A later calibration session gives a Hahn-echo time $T_2 = 16.42 \pm 0.35$ μs (Supplemental coherence-characterization section). Because the echo trace was acquired later, for a conservative transverse reference tied to the June 28 data we set $T_{2,\text{eff}} = T_2^* = 7.31$ μs, corresponding to $\Gamma_{f,T_2} = 1/(\pi T_{2,\text{eff}}) = 43.5$ kHz. The calculation in Fig. 2 uses these measurement-linked coherence parameters.

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior and maintains a narrow linewidth. However, even for pulse durations shorter

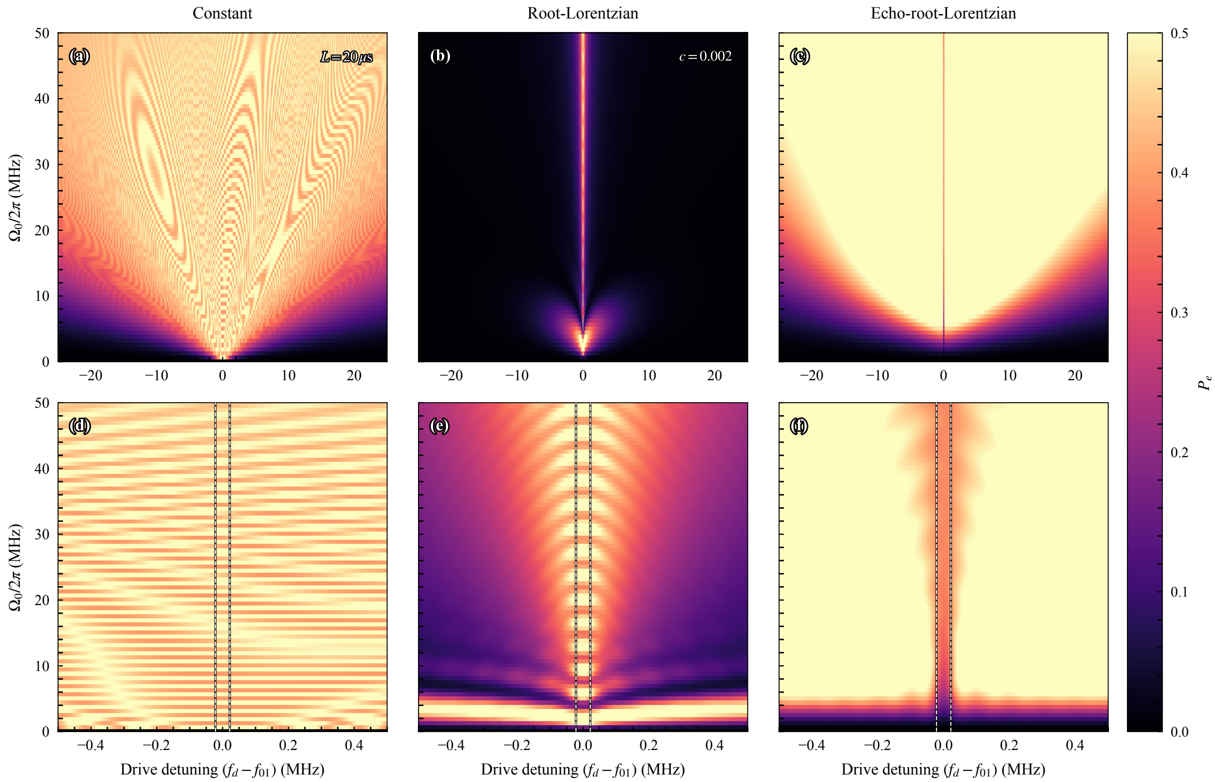

Figure 2. Three-state transmon simulation at two frequency scales for 20 μs pulses. The color scale is the final $|e\rangle$ population; the $|f\rangle$ population is evolved independently but is not added to P_e. The model uses anharmonicity $\alpha/2\pi = -200$ MHz. The upper row shows the broad 50 MHz domain (-25 to $+25$ MHz), and the lower row shows the narrow 1 MHz domain (-0.5 to $+0.5$ MHz). In the narrow panels, the paired black-edged white dashed lines at $\Delta/(2\pi) = \pm\Gamma_{f,T_2}/2$ enclose the coherence-limited FWHM; no reference line is drawn in the broad panels. From left to right, the columns show the constant, root-Lorentzian, and echo-root-Lorentzian protocols; the shaped pulses use relative endpoint amplitude $c = 0.002$. All six panels use the same coherence parameters, peak-Rabi grid, and probability scale. The comparison exposes conventional power broadening for the constant pulse, amplitude-dependent coherent structure for the ordinary root-Lorentzian pulse, and the narrow resonance-centered depletion feature preserved by the echo phase inversion.

than the qubit lifetime T_1, coherent oscillations arise for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase. For an ideal symmetric unitary two-level pulse, the corresponding forward and backward evolutions cancel exactly at zero detuning; relaxation, dephasing, waveform imbalance, and finite bandwidth make the cancellation incomplete in experiment. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. The waveform Fourier spectrum in Fig. 1 has a null at zero baseband frequency, consistent with its vanishing signed area, but this linear waveform property is not itself the nonlinear qubit response. The resonance-centered depletion instead follows from the driven dynamics and is supported by the state-evolution simulations and measurements. This mechanism can be viewed as a Loschmidt-echo-like evolution, in which ideal reversibility occurs only on resonance, while detuning, decoherence, and experimental imperfections break the cancellation and restore excitation [24, 25]. Combined with the adiabatic following of the Lorentzian envelope [12, 13], this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling narrow spectroscopy over a broad range of drive strengths with driven-spectroscopy pulses shorter than T_1. This sub-T_1 operating regime remains useful as qubit lifetimes increase because square and ordinary root-Lorentzian pulses can still exhibit coherent oscillatory distortions. Numerical state-evolution maps illustrating the resonant cancellation and detuned response are presented in the Supplemental Material [23].

Figure 3(a)–(f) resolves the two-dimensional maps into fixed-amplitude spectra. The three rows compare the same peak Rabi frequencies for the ordinary and echo protocols, making the echo-induced central depletion and

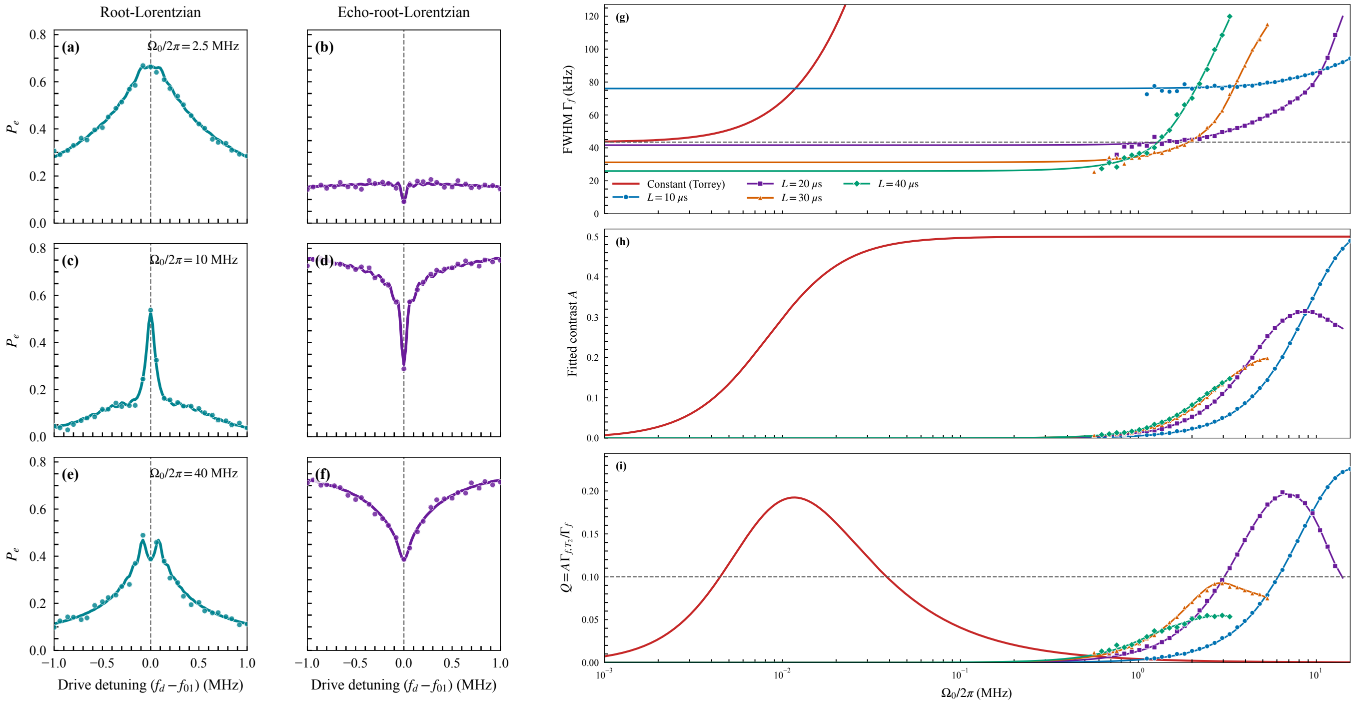

Figure 3. Spectra and resolution–contrast performance. (a)–(f) Final excited-state probability P_e versus detuning for 20 μs root-Lorentzian (left) and echo-root-Lorentzian (right) pulses with $c = 0.002$. From top to bottom, $\Omega_0/2\pi = 2.5$, 10, and 40 MHz. Solid curves are simulations, and markers are experimental data. Panels (g)–(i) show metrics versus peak Rabi frequency for echo-root-Lorentzian pulses with $L = 10$, 20, 30, and 40 μs. Colored continuous lines show simulations, and colored markers show experimental data. (g) Absolute fitted FWHM Γ_f in kHz, with the dashed line marking the coherence-limited reference $\Gamma_{f,T_2} = 43.5$ kHz; (h) fitted contrast A; and (i) contrast-weighted resolution $Q = AR = A\Gamma_{f,T_2}/\Gamma_f$. The red curve is the constant-drive Torrey result, and the dashed line in (i) marks $Q = 0.10$.

its persistence with increasing drive amplitude explicit.

To suppress residual power broadening at large drive amplitudes, the pulse envelope must satisfy requirements on its cutoff behavior. Truncation introduces a residual endpoint drive $\Omega_c = c\Omega_0$ and a corresponding cutoff-broadening contribution [13, 26, 27]. The relevant weak-saturation condition is $s_c = \Omega_c^2 T_1 T_2 \ll 1$, or $\Omega_c \ll 1/\sqrt{T_1 T_2}$; the inequality therefore points toward *smaller*, not larger, endpoint amplitudes. The relative cutoff c cannot be specified independently of Ω_0, T_1, and T_2. Our cutoff sweeps and high-amplitude measurements are reported in the Supplemental Material [23].

At the core of assessing pulse quality and identifying optimal operating regimes lies the spectral response of the driven qubit. We find that the echo pulse yields spectroscopic features with markedly reduced linewidths over a well-defined region of parameter space. We therefore choose pulse durations shorter than the qubit lifetime T_1 but long enough that Fourier broadening does not dominate.

Figure 3(g)–(i) quantifies this duration tradeoff at fixed cutoff $c = 0.002$. The continuous curves show the simulated response, while the markers show the experimental data.

Panels (g) and (h) separate spectral narrowing from signal loss: the longer pulses reach smaller absolute FWHM, but their fitted contrast remains lower over the usable drive range. Their contrast-weighted operating windows in panel (i) therefore contract despite the high resolution. Pulse duration must therefore be selected jointly with contrast and experimental signal quality, rather than by linewidth alone.

This constitutes an additional advantage of the echo protocol. While both constant and simple Lorentzian pulses exhibit coherent Rabi oscillations that persist even for pulse durations comparable to or exceeding T_1, the intrinsic π-phase inversion of the echo pulse suppresses these coherence-induced oscillations. As a result, the spectral response remains smooth and well-behaved, enabling reliable linewidth extraction in regimes where standard and Lorentzian pulses display pronounced oscillatory distortions; see the additional high-amplitude data in the Supplemental Material [23].

Finally, the use of formally infinite-support pulses introduces a three-dimensional design space comprising pulse duration, drive amplitude, and cutoff level. The corresponding cutoff–amplitude simulations and their resolution–signal analysis are reported in the Supplemental Material [23].

Amplitude robustness also requires that increasing the drive not displace the inferred resonance center. The measured plots span the full calibrated q1 sweep, $\Omega_0/(2\pi) = 0$–15.32 MHz. For the quantitative center summary, we use a predefined 42-slice high-Q subset. The selection uses finite-fit, contrast, and linewidth criteria but deliberately omits the usual center gate, so center stability is not enforced circularly.

Relative to their inverse-variance weighted mean, the fitted centers have an empirical RMS displacement of 3.02 kHz and their maximum absolute displacement is 7.47 kHz. A constant fit gives $\chi^2_\nu = 9.94$, so the local covariance errors do not provide a complete noise model and the data do not establish strict statistical constancy. Instead, these two empirical values bound the practical center excursion, including fitting variation, slow drift, and residual finite-pulse effects.

For comparison, the weak-drive three-level AC-Stark shift of a constant drive is $\delta f_{01} \simeq -(\Omega_0/2\pi)^2/[2(\alpha/2\pi)]$ [3, 28]. With $\alpha/(2\pi) = -200$ MHz, it reaches 0.20 MHz at the upper edge of the center-analysis subset. For the 10 μs root-Lorentzian envelope with $c = 0.002$, the phase-average shift is smaller by $\langle f^2\rangle_t = 0.003138$ and reaches only 0.63 kHz; the echo sign change does not cancel this quadratic term. Figure 4 displays this three-level scale separation through 60 MHz peak Rabi frequency. The full response maps and fine central-frequency scans are given in the Supplemental Material [23]. The measured few-kilohertz variations are nonmonotonic and show none of the approximately quadratic, hundreds-of-kilohertz displacement expected for a constant drive.

In conclusion, the echo-root-Lorentzian pulse combines adiabatic shaping with a π-phase inversion to provide high-resolution spectroscopy robust to drive amplitude. Simulations predict a narrow echo feature over more than three decades of peak drive. The measured sweep spans 0–15.32 MHz; within the predefined 42-slice center-analysis subset, the fitted center exhibits a 3 kHz RMS displacement and a 7.5 kHz maximum excursion. This makes the protocol well suited to precision qubit characterization and establishes phase-sensitive off-resonant response as a practical alternative to population-transfer spectroscopy.

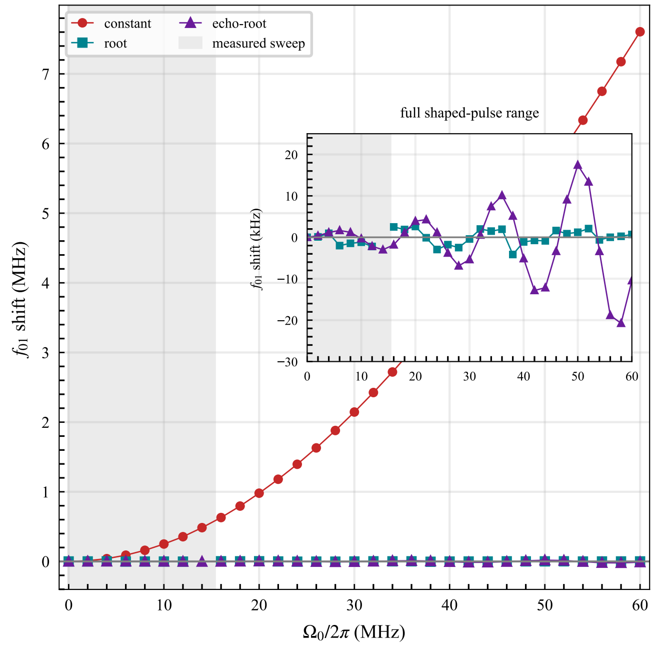

Figure 4. Drive-induced f_{01} displacement in the three-level model over an extended peak-Rabi-frequency range. The curves show the exact constant-pulse dressed shift and the spectral symmetry centers of the root-Lorentzian and echo-root-Lorentzian responses. The inset expands the shaped-pulse shifts over the complete 0–60 MHz simulated range; the gray band marks the measured 0–15.32 MHz sweep. One ambiguous root-pulse center outside the ±25 kHz display window is omitted. The finite shaped-pulse positions include coherent interference and are operational spectral centers rather than instantaneous dressed-state shifts.

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